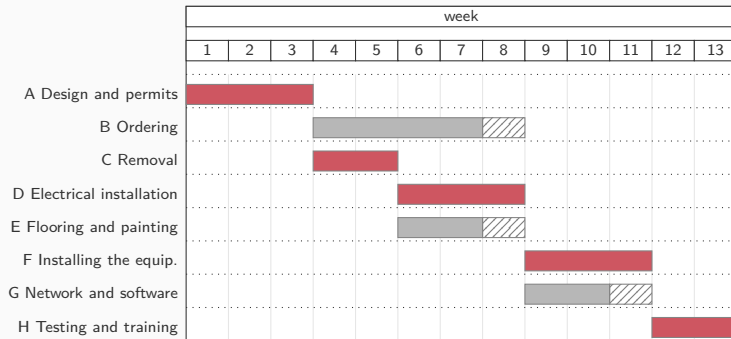


Complements of Operational Research

2.3 Gantt chart and project crashing

José Firmino Aguilar Madeira

From the plan to the calendar: the Gantt chart



Each activity is a bar, from its *ES* to its *EF* (dates from 2.1).

■ critical activities

■ activities with float

▨ float

Advantage: you can see at once who is working in each week.

Limitation: the precedences are not visible — that is why it is used **together with** the network, not instead of it.

What if the client wants the laboratory sooner?

Many activities can be **crashed** with more resources (overtime, more teams, express shipping), at a cost.

	normal		minimum		cost per week
	dur.	cost	dur.	cost	
A	3	6	2	9	3
B	4	20	2	26	3
C	2	4	1	6	2
D	3	8	2	12	4
E	2	5	2	5	—
F	3	10	1	18	4
G	2	6	1	9	3
H	2	4	1	9	5
	63				

durations in weeks; costs in thousands of euros

Cost per unit of time

$$\text{cost/week} = \frac{\text{minimum cost} - \text{normal cost}}{\text{normal dur.} - \text{minimum dur.}}$$

The cost is assumed to grow **linearly** between the normal and the minimum duration.

The client offers a **bonus of 6** (thousand euros) for each week gained relative to the 13.

Question: how many weeks is it worth cutting?

Rules for crashing

Step by step (one week at a time)

1. It is only worth shortening **critical** activities: shortening a non-critical one does not change the duration.
2. Shorten the **cheapest** critical activity (cost/week), without going below the minimum duration.
3. Recompute all paths.
4. With **several critical paths**, they must **all be shortened at the same time**: look for a common activity, or the cheapest combination (one per path).
5. Stop when the cost of one more week is greater than the bonus.

Careful with rule 4: this is where most errors happen. Shortening only one of the critical paths costs money and does not reduce the project duration.

That is why it helps to keep, at each step, a table with the duration of **all** paths.

Rule 5 with equality: if the cost of the next week is **equal** to the bonus, there is an economic tie — shortening or not gives the same total cost (this is what happens in Exercise 1 of Exercise sheet 2).

Crashing the laboratory, week by week

Path	normal	−C	−A	−F	−H	?
A−B−F−H	12	12	11	10	9	9
A−C−D−F−H	13	12	11	10	9	9
A−C−D−G−H	12	11	10	10	9	9
A−C−E−F−H	12	11	10	9	8	8
cost of the week		2	3	4	5	7

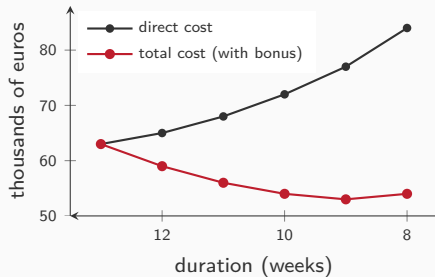
In red: critical paths at each step.

- **13→12**: a single critical path; the cheapest is C (2).
- **12→11**: two critical paths; A is common and costs 3.
- **11→10**: F is common to both and costs 4 (B+D would cost 7).
- **10→9**: three critical paths; H is common (5).
- **9→8**: A, C and H are already at their minimum. The cheapest is F+G or D+B, both costing 7. Since $7 > 6$ (bonus), **it does not pay off**: stop at 9 weeks.

Total cost as a function of the duration

Duration	shortened	direct cost	bonus	total
13	—	63	0	63
12	C	65	−6	59
11	A	68	−12	56
10	F	72	−18	54
9	H	77	−24	53
8	F+G (or D+B)	84	−30	54

Solution: finish in 9 weeks, shortening C, A, F and H by one week each; total cost 53 (against 63 without crashing).



The cost of each extra week **grows**; the bonus is constant. The optimum is where the cost of the next week exceeds the bonus.

Supplement: *crashing* as linear programming

Variables: s_j (start of activity j), y_j (weeks cut from j), T (project duration).

$$\min \sum_j c_j y_j - b(T_0 - T)$$

$$\text{s.t. } s_j \geq s_i + d_i - y_i \quad (i \text{ precedes } j)$$

$$T \geq s_i + d_i - y_i \quad (i \text{ final})$$

$$0 \leq y_j \leq d_j - m_j, \quad s_j \geq 0$$

d_j : normal duration; m_j : minimum duration; c_j : cost/week; b : bonus; T_0 : normal duration.

It is a **linear programming** problem (OR, 1st year): it is solved with the simplex method, `linprog` (MATLAB, Python) or the Excel Solver.

In the laboratory it gives the same result: $T = 9$, cost 53.

Advantage: it works for large networks, where the table of paths is no longer practical.

Exercise (exam, 26/07/2019)

Activity	Pred.	DN (days)	DM (days)	DN cost	DM cost
A	—	4	2	700	1100
B	—	9	7	900	1000
C	A	9	8	300	400
D	A, B	3	2	400	475
E	C, D	5	5	600	600
F	B, D	4	3	800	950

DN: normal duration; DM: minimum duration

- Determine the AON network, the float of each activity, the critical path, the project duration and its cost.
- Crash the plan to 16 days and determine the cost.
- Draw the Gantt chart of the final plan.

Watch out in b): at some point there are two critical paths with no common activity.

KEY TAKEAWAYS — 2.3

1. The **Gantt** chart shows the plan on the calendar ($ES-EF$, critical activities and floats); it is used **together with** the network.
2. **Cost per unit** = $(\text{min. cost} - \text{normal cost}) / (\text{normal dur.} - \text{min. dur.})$.
3. Shorten **the set of critical paths in the cheapest way**: one activity common to all, or the cheapest combination (one per path).
4. Keep the table of paths at each step; stop when the cost of the next week exceeds the bonus (if it is equal, both durations are equally optimal).
5. *Crashing* is a linear programming problem.

End of chapter 2. Next: 3 Scheduling problems (Johnson, TDM, *branch and bound*).

Exercise sheet 2 — Exercise 1 (homework)

Execution of an engineering project (1st test, 08/04/2019)

Activity	Precedence	DN (weeks)	DN cost (m.u.)	DM (weeks)	DM cost (m.u.)
A	—	3	10	1	20
B	A	2	25	1	30
C	A	5	40	4	55
D	B, C	7	35	5	45
E	D	7	15	5	35
F	E	2	15	1	25
G	D, F	4	20	3	45
H	G	2	30	2	30

- a) Determine the AON network, the possible time variation (float) of each activity, the critical path and the project duration.
- b) Given that the client offers a bonus of 10 m.u. for each week cut from the critical path, what is the best solution for the company? Give all the justifications that allow the critical path to be shortened and the company's profit to be increased.

Submission: by email to the lecturer, by the deadline given in class. All exercises: **Exercise sheet 2**.

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Project Management with PERT/CPM (*time-cost trade-offs*; linear programming formulation).
- Heizer, J., Render, B., *Operations Management*, Pearson — ch. 3 (Gantt charts, *crashing*).
- PMI, *PMBOK Guide*, 5th ed., 2013 — project time management.

Resources

- Exercise sheet 2. Practical assignment no. 1: create and solve 2 PERT and CPM problems with *crashing*.